2190 Final Report

2190 was made to drill Lower Marmaton Laterals in the Anadarko Basin, Texas Panhandle for Henley & Associates (Quanah Panhandle). Carbonate formation just above the granite wash. Lateral, in at 7,015', out at 11,900'. Bent Motor, .64 rev/gal (@275 gal/min → Total rpm = 240). Knuckles in cone (80'/hr @ 240rpm). End goal is to change the interval to 1 or 2 bits and maintain (if not improve) ROP.

2 bit bodies have been built to date and run 3 times each. Each well drilled has used 2 bits each; however, the dulls have looked very good. The bits have been pulled for motor failures or MWD failure a total of 5 times in the 3 wells. On two occasions the motor was changed, and they reran the bit. From the dulls, there is no indication that the wells could not be done in one bit every time, but the current limiting factor is the motors. The ROP of the design was an improvement over the previous bits as well. As can be seen in figure 1, the 2190 has saved 2.5 to 2.75 days on each well.

The goal was to get a data point to go off in a new market, and the 2190 has become the lateral bit of choice for Quanah Panhandle. It lowered the number of bits required to 2 bits (with indications that one bit will successfully complete the interval when/if Quanah acquires a better motor) and increased the ROP.

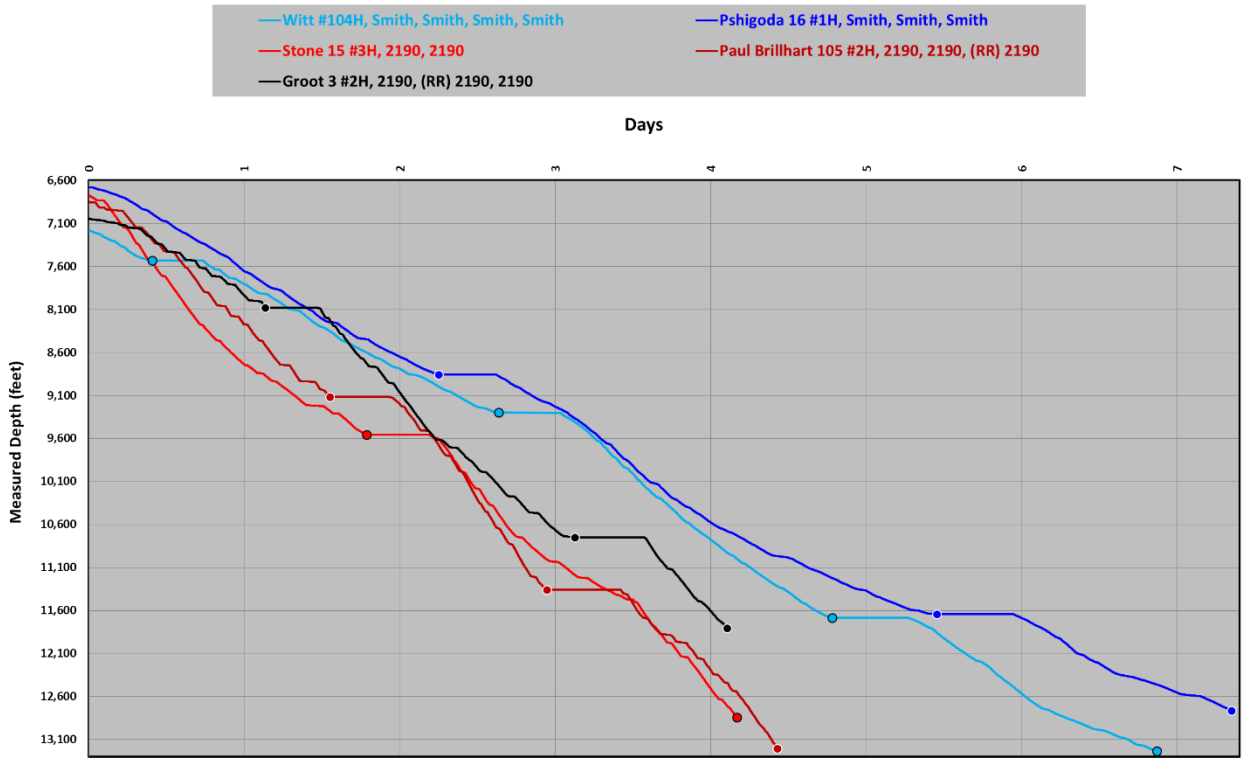

Figure 1: Depth vs. Days plot for the three wells drilled with the 2190 (the two red curves and the black curve) and two wells drilled with Smith bits (the 2 blue curves).

See figures 2 and 3 for financial implications on the design.

Bit Description (Assembly if known)		*Leave this field blank if a new bit design*
Bit Size (inches)	6 1/8	
Blade Count & Cutter Size	516	
Cutter Placement	SRK	
Tool Joint	Short	S
Gauge	Short	S
Application	Medium	
Cutter Type	DP	DP=Double Pressed, SL=Shallow Leached, NL=Non-Leached
Projected Sales Price	$ 7,500	
Customer Average Days to Pay	30	
Bit Utilization (Total Runs/Month)	2.00	*Includes Clean & Paints*
Clean & Paints per month	-	
Turnkey Runs per Invoice	-	*If every 4th Invoice needs additional bit; input ".25"*
Bit Life (# Runs including C&P and Turnkeys)	6.00	*If Unknown Estimate 6 Runs*
Total ROI per Bit	56.62%	
Gross Profit Margin	53.16%	
Gross Return on Investment	113.48%	
Diamond Repair Costs as % of Revenue	18.20%	
Repair Costs as a % of Revenue	21.57%	

Figure 2:ROI form as filled out by sales with the design request.

Bit Description (Assembly if known)	2190	*Leave this field blank if a new bit design*
Bit Size (inches)	6 1/8	
Blade Count & Cutter Size	516	
Cutter Placement	SRK	
Tool Joint	Short	S
Gauge	Short	S
Application	Easy	
Cutter Type	DP	DP=Double Pressed, SL=Shallow Leached, NL=Non-Leached
Projected Sales Price	$ 7,500	
Customer Average Days to Pay	30	
Bit Utilization (Total Runs/Month)	1.50	*Includes Clean & Paints*
Clean & Paints per month	-	
Turnkey Runs per Invoice	-	*If every 4th Invoice needs additional bit; input ".25"*
Bit Life (# Runs including C&P and Turnkeys)	4.00	*If Unknown Estimate 6 Runs*
Total ROI per Bit	55.33%	
Gross Profit Margin	58.14%	
Gross Return on Investment	138.86%	
Diamond Repair Costs as % of Revenue	18.20%	
Repair Costs as a % of Revenue	22.70%	

Figure 3: ROI form filled out after three wells drilled, 6 runs on 2190.

The final recommendation from engineering on 2190 is to change the status from test to production.